LifeGrid — Live demo script

One story, start to finish: Southfield General Hospital is running out of O+ blood. A manager appeals to donors and a donor answers from their phone. Then a family requests blood for a patient, the hospital confirms it, and the network covers it from stock. Finally the nightly plan, a new donor, and the audit trail. About 35–45 minutes.

Before the demo (5 minutes)

0.1 **Reset the demo data** (only needed if it has been used since the last reset). In the Neon console open **SQL Editor** and run `DROP SCHEMA public CASCADE; CREATE SCHEMA public;`. Then in a terminal, in `lifegrid\backend`, run `.venv\Scripts\python scripts\seed_remote.py` and paste the Neon connection string. This rebuilds the clean demo region for today's date.

0.2 Open `https://lifegrid01.vercel.app` a few minutes early and click once, so the server and database are awake.

0.3 Every staff account's password is `synthetic-demo-pass`. On the sign-in page, **Demo accounts** fills the form for you. Only the administrator needs a 6-digit code: add the setup key `LIFEGRIDDEMOSECRETKEYAAAAAAAAAAA` (time-based) to Google Authenticator or any authenticator app once.

0.4 Phone app sign-in uses a one-time code. No SMS is sent in the demo: the code appears on screen under **Development code**; tap it to fill it in.

0.5 One browser holds one sign-in. Use a normal window for staff and a phone (or a private window in phone view, `F12` then `Ctrl+Shift+M`) for the donor and requester, side by side. Sign out is at the bottom of the menu.

0.6 To run the demo again later, repeat step 0.1.

The cast

Sign in with	Who	Role in the story	Code?
`nabila.karim@lifegrid.test`	Nabila Karim	Blood bank manager. Sees the whole region, starts the donor appeal, approves and dispatches transfers.	No
`tanvir.rahman@lifegrid.test`	Tanvir Rahman	Transfusion lead at Southfield General Hospital (H-01). Confirms requests, receives transfers, records donations.	No
`imran.chowdhury@lifegrid.test`	Imran Chowdhury	Donor coordinator. Sees donors by reference only; verifies groups and records deferrals.	No
`sadia.haque@lifegrid.test`	Sadia Haque	Administrator. Changes a rule and shows its history.	Yes
`mahmud.alam@lifegrid.test`	Mahmud Alam	Auditor. Read-only; verifies the audit chain at the end.	No

`+8801000000001`	The donor	O+, lives about 1 km from Southfield, eligible and has never donated. Phone app.	On screen
`+8801000009999`	The requester	A relative asking for blood for a patient at Southfield. Phone app.	On screen
`+8801000005555`	A new donor	Not in the system yet; registers in Part 6.	On screen

What's in the demo data: 14 sites: two regional blood banks (**Riverside** and **Lakeview**) and twelve hospitals. About 1,230 red-cell, platelet and plasma units, 8 weeks of usage history, 400 synthetic donors, and a transfer plan that ran "last night". **Southfield General Hospital has only 2 O+ red-cell units** while its forecast says it will use 7 in the next 2 days. Everything is synthetic.

PART 1 · The manager's morning

Nabila Karim, blood bank manager

Sign in: `nabila.karim@lifegrid.test` no code needed

1.1 **Network overview** opens. Point out the key figures: **Red cells available**, **Expiring in 2 days**, **Projected stockouts** and **Open alerts**.

1.2 **Sites by risk**: every hospital and blood bank on the map, coloured **Critical**, **Watch** or **Stable**. Click Southfield to show its counts and the **Open site stock** link.

1.3 **Available stock by site and group**: on the Southfield row the **O+** cell shows **2**. The small bar under each number shows how much expires within 2 days (red), in 3–7 days (amber) or later. Switch to **Platelets** at the top right: they last only 5 days, so expiry pressure is much higher.

1.4 The **Low stock** panel at the top lists **O+ · Southfield General Hospital** first: **2 on hand · 7 needed in 2 days**. This comes from the forecast, before anyone has run out.

1.5 Open **Forecasts**. Choose Southfield General Hospital, Red cells, O+. Black line: the last 28 days of real use. Dashed line: the next 7 days. Shaded band: the likely range. The right panel names the model picked for this series and compares its error with the simple "same day last week" baseline.

What this shows: the whole region on one screen, and a warning that comes from a forecast, days before the shelf is empty.

PART 2 · An appeal reaches a donor's phone

Nabila (staff window) and the donor (phone) — Phone: `+8801000000001` code on screen

2.1 Back on the overview, in **Low stock**, press **Appeal to donors** on the O+ Southfield row. The dialog **Appeal for O+ at Southfield General Hospital** has **Units needed** filled in (projected shortfall **5**) and urgency **Urgent**. Press **Send appeal**.

2.2 The message reads **Appeal AP-… started. 7 donors invited in the first wave.** (7 in the daytime; fewer at night, when some donors' quiet hours apply.) The row now says **Appeal running**. Open **Requests**: the appeal is listed as **Donor appeal**. Open it to show the 7 invited donors in wave 1, by reference and blood group only, without names or phone numbers.

2.3 **On the phone**, open `lifegrid01.vercel.app`. Tap **I want to donate**, enter `+8801000000001`, **Send code**, tap the development code, **Continue**. Home shows **You can donate** and a red bar: **1 open invitation**.

2.4 Tap it. The card shows **O+**, **Hospital in Southfield**, **Urgent**, when it is needed, and a **Stock appeal** label. The donor sees the area, not the hospital's name. Press **I can donate**. The card changes to **Accepted**.

2.5 Press **Open messages**. Type "I can come at 4 pm today." and press **Send message**. Point out **Phone numbers: Not shared**.

2.6 **Staff window**: refresh the appeal. The donor is now **Accepted**. Press **Message** next to them and reply "Thank you! Please come to the blood bank desk, ground floor." The reply appears on the phone.

2.7 Optional: press **Share my number** on both sides. Numbers appear only once both have agreed. Then sign out of the staff window.

What this shows: LifeGrid invited only donors who are compatible, eligible, available, nearby and outside their quiet hours, and only a few at a time: enough to expect the shortfall plus 50% to accept. Donor and hospital talk without exposing anyone's number.

PART 3 · The hospital records the donation

Tanvir Rahman, transfusion lead, Southfield — Sign in: `tanvir.rahman@lifegrid.test` no code needed

3.1 Tanvir lands on **Site stock** for Southfield General: only his hospital's units, first-expired first.

3.2 Open **Requests**, then the appeal **AP-…**. In the donor list press **Donated** next to the donor who accepted. The appeal shows **1 of 5 secured**.

3.3 **On the phone**, go to **Home**: **Eligible again in 120 days**, with the date. The donor won't be invited again until then.

3.4 Sign out of the phone app (**Sign out** in its menu). Keep Tanvir signed in for Part 4.

What this shows: the outcome updates the appeal, starts the donor's waiting period and feeds their reliability score. Only real outcomes change reliability; declining an invitation never counts against anyone.

PART 4 · A family asks for blood

The requester (phone), then Tanvir — Phone: `+8801000009999` code on screen

4.1 **On the phone**: tap **I need blood for a patient**, enter `+8801000009999`, tap the code, **Continue**. **My requests** is empty. Tap **New request**.

4.2 Patient blood group **B−**, component **Red cells**, units needed **6**, hospital **Southfield General Hospital**, urgency **Urgent**, needed by **this evening**. Press **Submit request**. Status: **Awaiting hospital**: stock and donors stay untouched until the hospital confirms.

4.3 **Tanvir**: **Requests** shows **1 request needs confirmation** at the top. Open **RQ-…**, check it against the patient's record, and press **Confirm request**.

4.4 Result: **Covered by stock** and the message **Stock covers this request. No donors were contacted.** LifeGrid reserved the **5** compatible units at Southfield (2 B− and 3 O−; a B− patient can receive only B− or O−) and proposed **1** more from Westwood District Hospital, 13 km away. A link appears: **Review the proposed transfers for this request**.

4.5 **On the phone** the status updates by itself: **Covered by stock**, units secured, **6 from stock**, **0 donors notified**, and the four-step progress bar.

What this shows: a hospital confirms every request, and LifeGrid looks at stock across the whole network before it asks a single donor. Compare Part 2, where stock was the problem and donors were the answer.

PART 5 · Moving units across the network

Nabila approves and dispatches; Tanvir receives and issues — Sign in: `nabila.karim@…`, then `tanvir.rahman@…`

5.1 **Nabila**: **Requests** → the request **RQ-…** → **Review the proposed transfers for this request**. One line: **Westwood District Hospital to Southfield General Hospital**, B−, 1 unit, **For a blood request**. Press **Approve** and confirm **Approve transfer of 1 unit?**

5.2 **Transfers** → **To dispatch** → **Dispatch**. LifeGrid picks the actual unit, first-expired-first-out, from units with enough shelf life left to travel (5 days for red cells). Press **Dispatch**. It moves to **In transit**: **Awaiting receipt at destination**.

5.3 Still Nabila: open **Transfer plan** (last night's plan). **Projected effect** compares expired and short units without and with the plan. Find **Lakeview Regional Blood Bank to Southfield General Hospital**, O+, **13 units**, **Projected shortage at destination**: the network's answer to the same O+ shortage. Lower it to **10** and press **Approve**. The dialog notes **You reduced this transfer from 13 to 10 units.** Optional: **Reject** a **Relay or swap between sites** line, to show nothing moves without a person. Sign out.

5.4 **Tanvir**: **Transfers** → **In transit** → **Receive** on the unit from Westwood, and confirm. The unit is now available at Southfield, ready for the patient's sixth unit.

5.5 **Stock**: filter status **Reserved**. The 5 units held for the request are listed. Tick them, press **Issue** in the bar at the bottom, and confirm **Issue 5 units?** Click one unit code: its history shows every step with who did it, when and why. Sign out.

What this shows: the optimizer weighs waste, shortages and distance across the region, and a person approves every move. Units travel first-expired-first-out, and a unit's location changes only when the destination receives it.

PART 6 · A new donor joins

A new person, on the phone

Phone: `+8801000005555` code on screen

6.1 Tap **I want to donate**, enter `+8801000005555`, tap the code, **Continue**. Because this number is new, **Set up your donor profile** opens.

6.2 Pick blood group **A+**. Tap the map near Southfield: the pin is rounded to about 1 km. Birth year **1998**, tick **I weigh at least 50 kg**, read and tick the consent, then press **Save and continue**. Home says **You can donate**.

6.3 Set **Quiet hours** from 22:00 to 07:00 and tick **Still contact me during quiet hours for emergencies**. Set **Availability** to **Travelling** to pause invitations, then back to **Available**. Changes save as you make them.

6.4 Open **My data**: what LifeGrid holds (no name, no address, no medical history), **Download my data** and **Delete my data**. Don't delete during the demo.

What this shows: privacy by design. Phone numbers are encrypted, the area is rounded, and the donor can see, export or erase their data at any time.

PART 7 · The donor coordinator

Imran Chowdhury

Sign in: `imran.chowdhury@lifegrid.test` no code needed

7.1 **Donors** opens with the Low stock panel on top: Southfield O+ now shows **Appeal running**. The heading says **Contact details are never shown here**.

7.2 Filter group **A+** to find the new donor. The dashed group chip means **Self-reported**. Press **Verify group**: the chip turns solid, **Group verified**.

7.3 On another donor press **Defer**, pick a category and an **Eligible again on** date, then **Save deferral**; LifeGrid keeps just the category and date. Sign out.

What this shows: the people who manage donors work with references and rules, never personal details.

PART 8 · Rules and the audit trail

Sadia Haque (administrator), then Mahmud Alam (auditor) Sign in: `sadia.haque@…` authenticator code: yes

8.1 **Sadia**: password, then the 6-digit code from the authenticator app, then **Verify and continue**. Open **Administration** → **Settings**.

8.2 Change `max_invites_per_30d` from 2 to **3** and press **Save**. **Change history** on the right shows the old value, the new value, who changed it and when. Rules like shelf life, waiting periods, wave timing and search distances all live here, not in the code.

8.3 **Compatibility** tab: rows are patients, columns are donor groups, numbers are preference. O− comes last for everyone who isn't O−, so it's kept for the patients who need it. **Users** tab: role, site scope, authenticator and the **Active** switch. Sign out.

8.4 **Mahmud** (`mahmud.alam@lifegrid.test`, no code): **Audit log** lists every action from today: the appeal, the request and its confirmation, the transfers, the issued units, the settings change, every sign-in. Press **Verify chain**: **Chain intact. N events verified.**

What this shows: every sensitive action is chained to the one before it by a cryptographic hash. If anyone edited a past record, Verify chain names the first event that no longer matches. Auditors can check everything and change nothing.

How it all connects

When this happens	It shows up here
A hospital's stock falls below its forecast	Low stock on the Overview and Donors screens, Projected stockouts, and a transfer in the nightly plan
A manager sends an appeal	An SMS (mock) and an invitation on eligible donors' phones; a Donor appeal on Requests with the invited donors
A donor accepts	Accepted on the appeal; a private conversation with the hospital team
The hospital records Donated	Units secured on the appeal; the donor's 120-day waiting period and reliability
A requester submits a request	Needs confirmation on the hospital lead's Requests screen. Nothing else happens yet
The hospital lead confirms	Units reserved at the hospital, transfers proposed from other sites, status on the requester's phone; donors only for a shortfall
A manager approves and dispatches	Transfers: In transit at both ends; the units' history
The destination receives	Units available at the new site; the stock matrix on the Overview
An administrator changes a rule	Change history; LifeGrid uses the new value from that moment
Any of the above	The hash-chained audit log, checked with Verify chain

LifeGrid only recommends: a person approves every transfer, a hospital confirms every request, and donations happen only at licensed facilities. Clinical values in the demo are placeholders that need clinician review before real use.